



HK IMO Training 2021-2022
Problem Set 1
For 25 September 2021



Problem 1. The integers from 1 to 100 are written in a row in some order. A number is *good* if it is greater than all numbers to the right of it. A number is *nice* if it is greater than all numbers to the left of it. Suppose there are exactly k good numbers and exactly k nice numbers. Find the largest possible value of k .

Problem 2. For each positive integer n , let d_n be the largest integer that divides $a^n + (a+1)^n + (a+2)^n$ for every positive integer a .

- (a) Prove that for every positive integer n , the number d_n is of the form 3^k for some nonnegative integer k .
- (b) Prove that for every nonnegative integer k , there exists a positive integer n such that $d_n = 3^k$.

Problem 3. Let O be the centre of a circle Γ . Let A be a point on Γ with tangent ℓ . A line parallel to ℓ cuts Γ in two points B and C . The line CO intersects ℓ at D . Let E be the intersection point of the lines AO and DB . Prove that the line passing through C and parallel to the tangent at O to the circumcircle of $\triangle OCE$ intersects the line BD on Γ .

Problem 4. Find all polynomials $P(x)$ with real coefficients such that

$$P(\sqrt{2}x) = P(x + \sqrt{1-x^2})$$

for all real number x with $|x| \leq 1$.